

WIPRO CAPSTONE PROJECT REPORT

Name	Nitin Yadav
Superset ID	5388156
Organization	Wipro
Project Type	Capstone Project
Project Name	Automation Testing using Selenium PyTest Framework
	Automation Testing using Behave BDD Framework
Domain	E- commerce (Myntra)
Module Title	Men Section

SYNOPSIS

The Wipro Capstone Project focuses on automating the Myntra MEN module using two separate automation frameworks: Selenium WebDriver with Python PyTest framework and Behave BDD framework. The automation validates product search, category navigation, product selection, size selection, add to bag functionality, wishlist functionality, filter application, sort functionality, donation selection, and order placement workflow.

The framework was designed using the Page Object Model (POM) architecture to improve maintainability, scalability, and reusability of automation scripts.

Screenshots, logs, Allure reports, and reusable utility modules were integrated to support execution tracking, debugging, and reporting.

The project includes both positive and negative testing scenarios using industry-standard automation testing practices. Data-driven testing was also implemented using CSV files to execute multiple test datasets efficiently.

PROJECT DESCRIPTION

The automation framework automates the end-to-end workflow of the Myntra MEN shopping module. The framework performs MEN category navigation, casual shoes selection, product search validation, size selection, add to cart functionality, wishlist validation, filter application, sorting validation, donation amount selection, and order placement verification.

Separate page classes were implemented for MEN module functionalities using Selenium Python Page Object Model (POM) framework. Utility modules were

created for driver setup, screenshots, logging, and test data handling to improve code reusability and framework maintainability.

SELENIUM PYTEST FRAMEWORK

TEST STRATEGY

The project uses Selenium WebDriver with Python PyTest framework for automation testing. The framework supports:

- End-to-End Testing
- Positive Testing
- Negative Testing
- UI Functional Testing
- Data-Driven Testing

Tools Used:

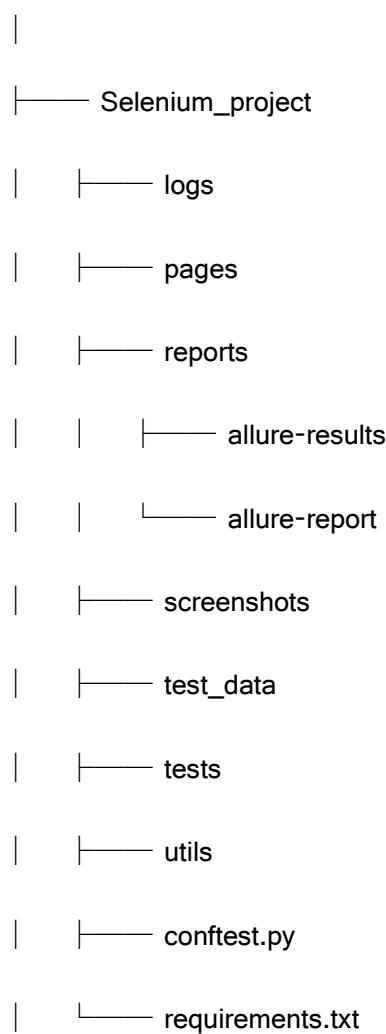
- Selenium WebDriver
- Python
- PyTest
- Behave BDD
- Allure Reports
- Logging Module
- CSV Data Handling
- Page Object Model (POM)
- WebDriver Manager

TEST ENVIRONMENT

Component	Details
Operating System	Windows 11
IDE	PyCharm
Browser	Google Chrome
Framework	Selenium WebDriver + PyTest
Programming Language	Python
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports
Test Data Handling	CSV
Version Control	Git & GitHub

FRAMEWORK ARCHITECTURE

Wipro_Myntra_Men_Capstone_Project



The automation suite includes:

- 1 End-to-End Test Case
- 4 Positive Test Cases
- 2 Negative Test Cases

The End-to-End test validates the complete Myntra MEN shopping workflow from MEN category navigation until Place Order page verification.

SELENIUM TEST CASE SUMMARY

S.No	Test Type	Test Scenario	Expected Result	Status
1	End-to-End	Complete MEN Shopping Workflow	User successfully reaches Place Order/Login page	Passed
2	Positive	Search Functionality Validation	Search results displayed successfully	Passed
3	Positive	Filter Functionality Validation	Puma filter applied successfully	Passed
4	Positive	Sort Functionality Validation	Better Discount sorting applied successfully	Passed
5	Positive	Wishlist Functionality Validation	Wishlist/Login popup displayed successfully	Passed

6	Negative	Invalid Search Validation	No result message displayed successfully	Passed
7	Negative	No Size Selected Validation	Size selection error message displayed successfully	Passed

END-TO-END FLOW VALIDATION

The End-to-End automation validates the complete business workflow including:

- Website Launch
- MEN Menu Navigation
- Casual Shoes Category Selection
- Product Selection
- Size Selection
- Add to Bag Functionality
- Cart Validation
- Donation Amount Selection
- Total Amount Verification
- Place Order Navigation
- Login Page Redirection Verification

POSITIVE TEST CASES

Positive Test Case 1 – Product Search Functionality

- Validates product search functionality
- Searches valid product “Sneakers”
- Verifies search result page display
- Ensures matching products are displayed successfully

Positive Test Case 2 – Brand Filter Validation

- Validates brand filter functionality
- Applies “Puma” brand filter
- Verifies filter selection successfully
- Ensures filtered products are displayed

Positive Test Case 3 – Product Sorting Validation

- Validates sorting functionality
- Opens sorting options
- Selects “Better Discount” option
- Verifies products are sorted successfully

Positive Test Case 4 – Wishlist Functionality Validation

- Validates wishlist functionality
- Opens selected product
- Clicks Wishlist button
- Verifies login popup display successfully

NEGATIVE TEST CASES

Negative Test Case 1 – Invalid Product Search

- Searches invalid product value
- Validates application handling for invalid search
- Verifies no valid matching products are displayed
- Ensures invalid product search handling works correctly

Negative Test Case 2 – Add To Bag Without Selecting Size

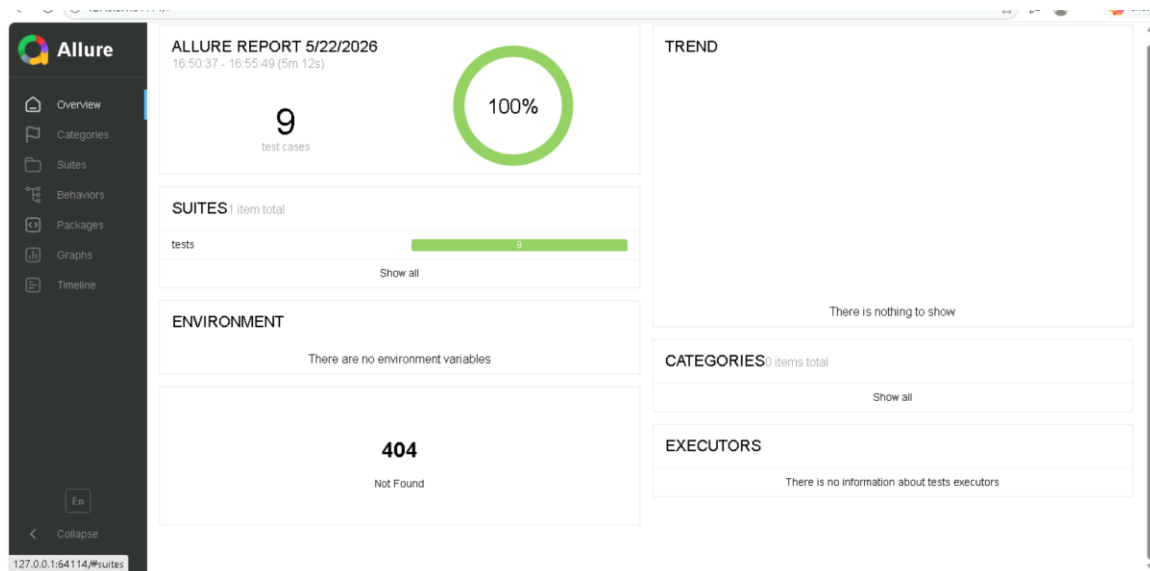
- Opens product page
- Clicks “Add To Bag” without selecting size
- Validates application error handling
- Verifies display of:

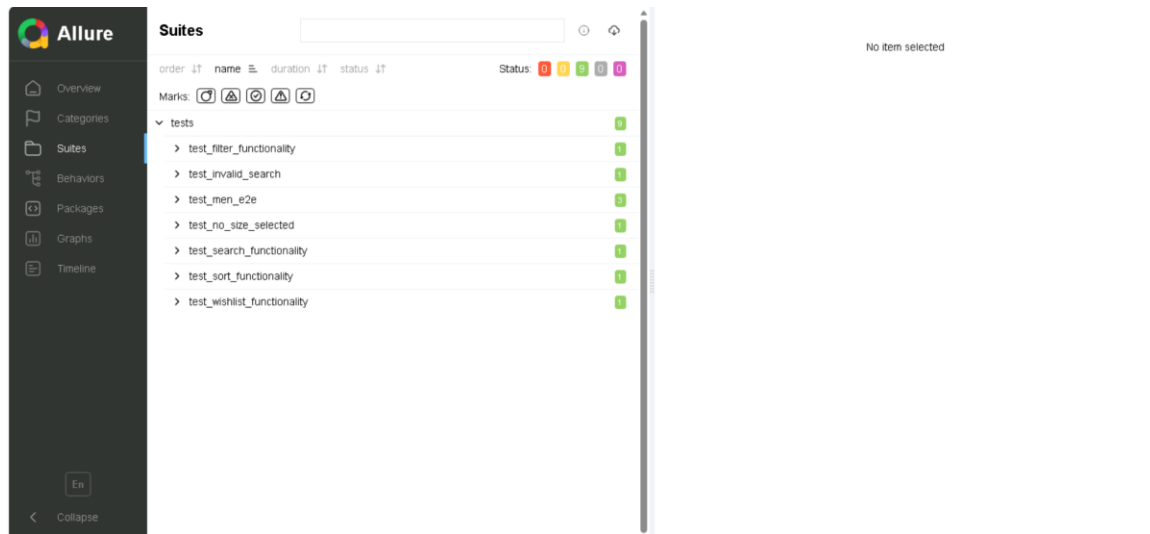
“Please select a size”

- Ensures product cannot be added without mandatory size selection

ALLURE REPORTS

- Integrated Allure Reports with PyTest
- Generates professional execution reports
- Displays passed and failed test status
- Captures screenshots during execution
- Maintains execution history
- Displays step-by-step execution flow
- Helps debugging and tracking failures
- Provides detailed execution summary





TEST OUTPUTS

The framework automatically captures screenshots during important execution stages including MEN category navigation, product selection page, filter application page, cart page, wishlist popup, donation selection page, and place order/login verification page.

Detailed execution logs are generated using the Python Logging framework for debugging, execution tracking, and failure analysis. Logs and screenshots are attached directly to Allure Reports for better traceability and reporting.

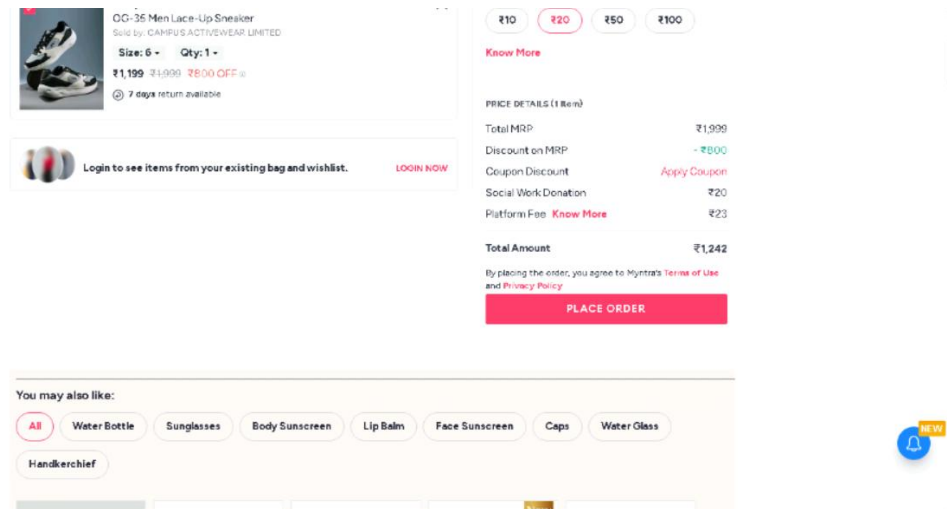


Fig 1: E2E

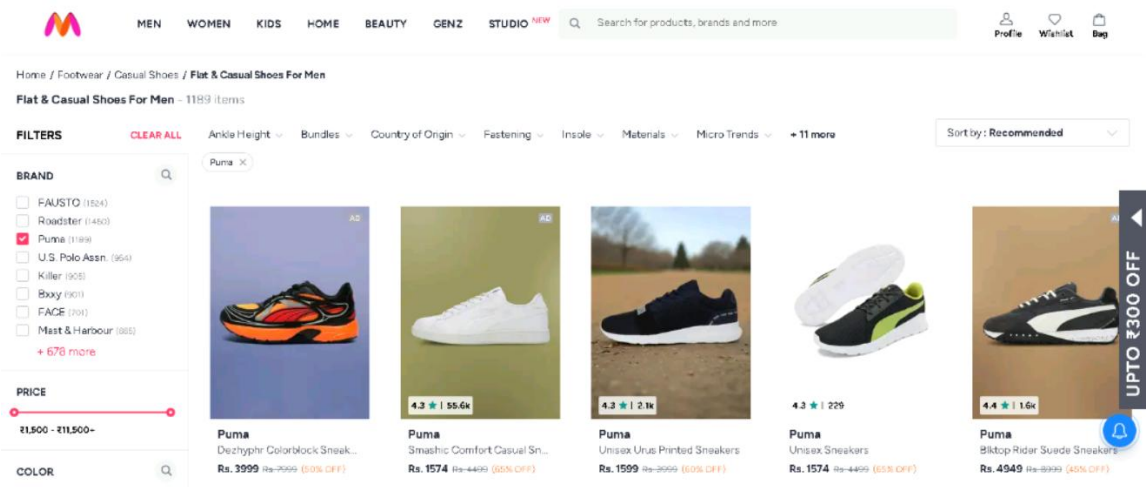


Fig 2: Filter

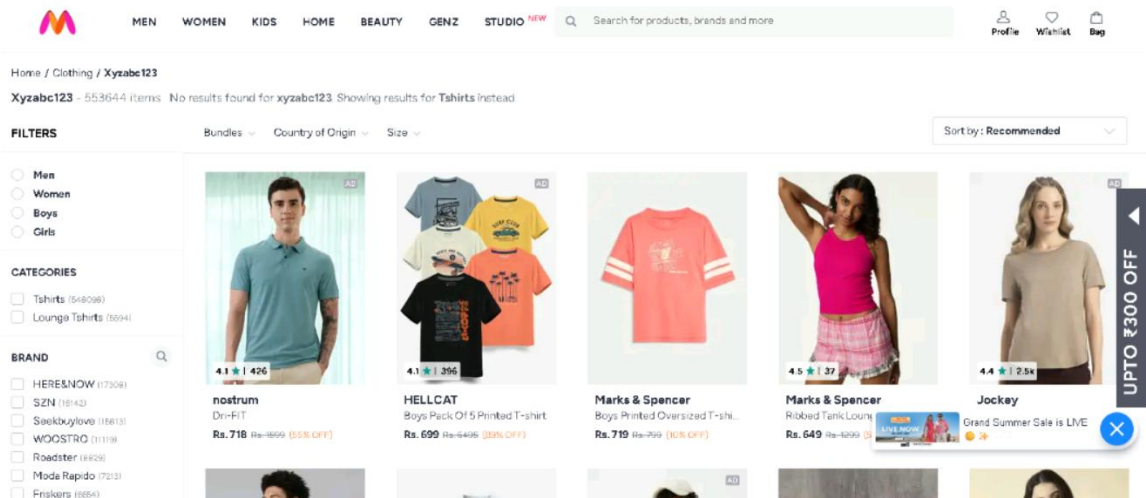


Fig 3: Invalid Search

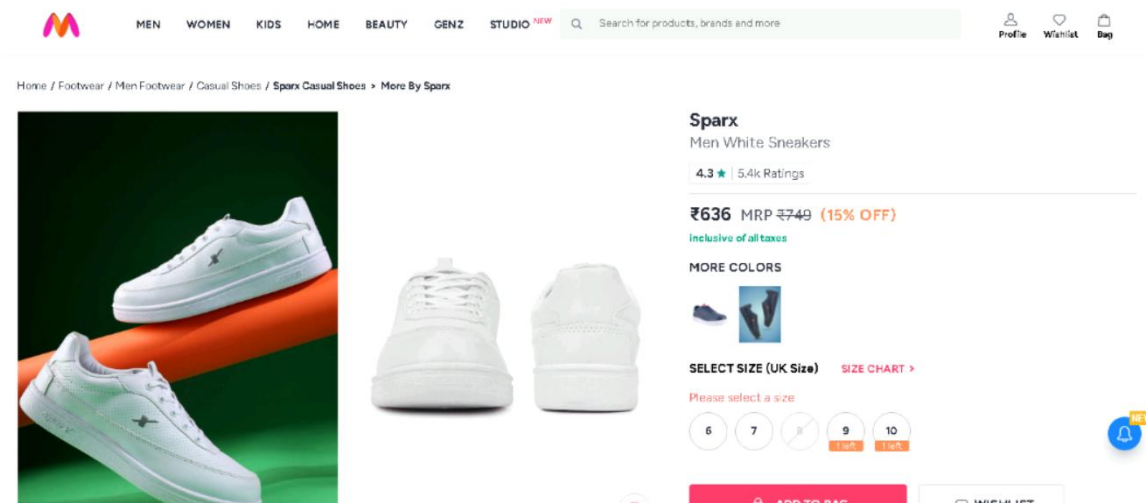


Fig 4: No Size selected

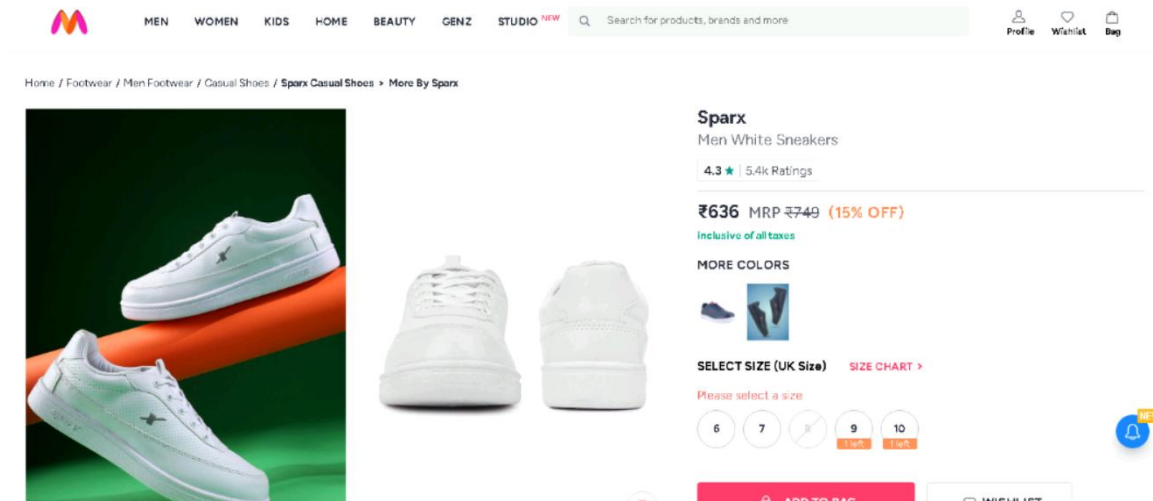


Fig 5: Search

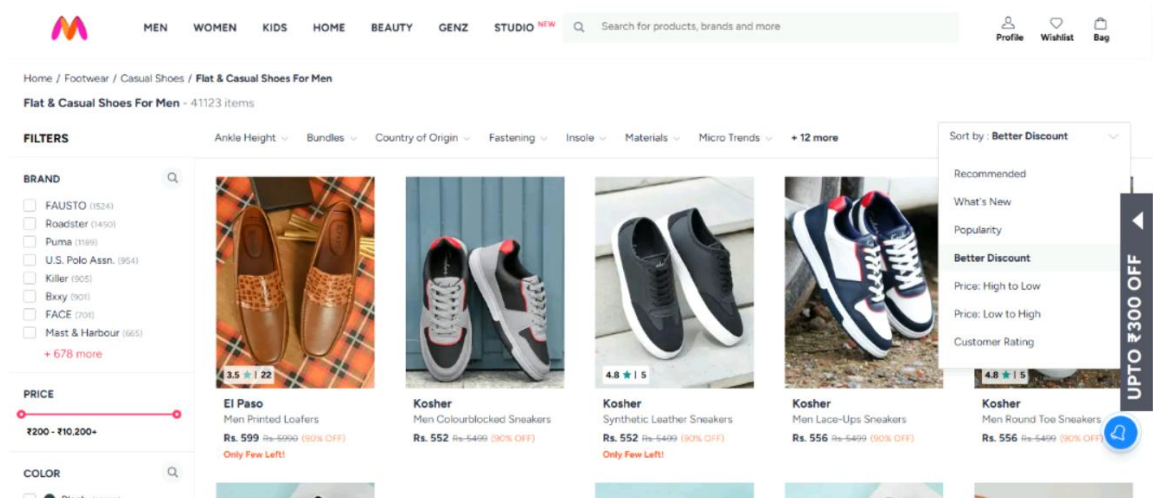


Fig 6: Sort

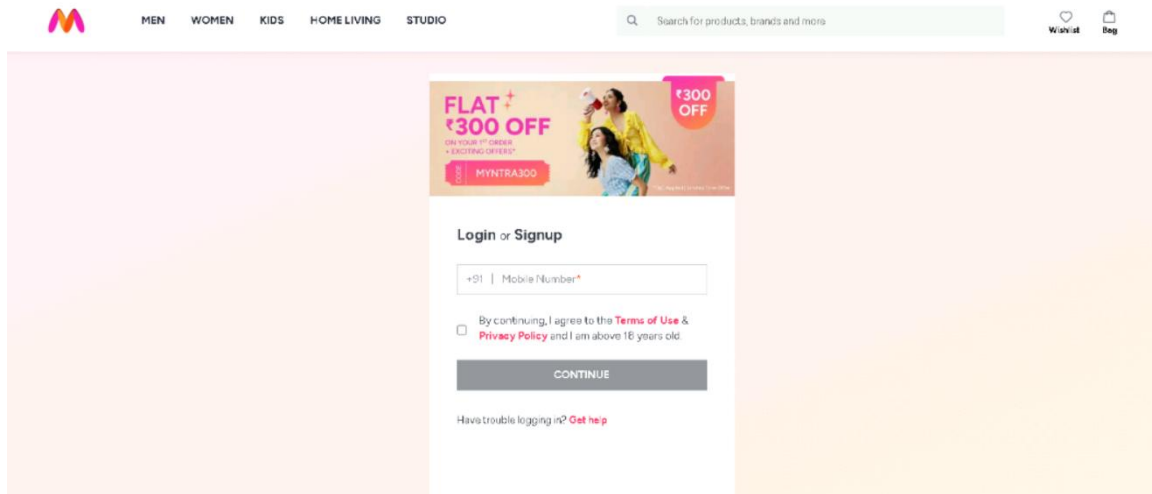


Fig 7: Wishlist

TEST REPORT

- ✓ Total Tests Executed: 9
- ✓ Tests Passed: 9
- ✓ Tests Failed: 0
- ✓ Automation Accuracy: 100%

CONCLUSION

The Selenium PyTest automation framework successfully validates the Myntra MEN module using End-to-End, Positive, and Negative testing approaches.

The framework is designed using:

- Selenium WebDriver
- PyTest Framework

- Page Object Model (POM)
- Allure Reporting
- Logging Framework
- Data-Driven Testing

The automation framework improves:

- Test Maintainability
- Reusability
- Scalability
- Reporting Quality
- Automation Reliability

The framework successfully validates critical business workflows including MEN menu navigation, product search functionality, filter validation, product sorting, wishlist functionality, cart validation, donation selection, and place order verification.

All test cases executed successfully with professional reporting, screenshot capture, and detailed execution logs.

Overall, the project demonstrates a practical and professional Selenium automation framework for real-world e-commerce web application testing.

BEHAVE BDD FRAMEWORK

TEST STRATEGY

The project uses Behave BDD framework integrated with Selenium WebDriver for automation testing.

The framework supports:

- End-to-End Testing
- Positive Testing
- Negative Testing
- Behavior Driven Development (BDD)
- UI Functional Testing

TOOLS USED

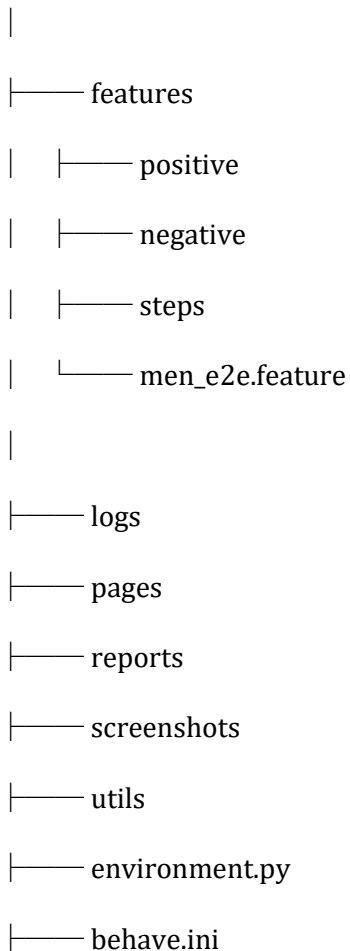
- Selenium WebDriver
- Python
- Behave BDD
- Gherkin Syntax
- Allure Reports
- Logging Module
- Page Object Model (POM)
- WebDriver Manager
- Git & GitHub

TEST ENVIRONMENT

Component	Details
Operating System	Windows 11
IDE	PyCharm
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Programming Language	Python
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports
Test Data Handling	CSV
Version Control	Git & GitHub

FRAMEWORK ARCHITECTURE

Wipro_Myntra_BDD_Project



└── requirements.txt

TEST EXECUTION

The automation suite includes:

- 1 End-to-End Test Case
- 4 Positive Test Cases
- 2 Negative Test Cases

The End-to-End test validates the complete Myntra MEN shopping workflow from MEN category navigation until Place Order login redirection validation.

SELENIUM TEST CASE SUMMARY

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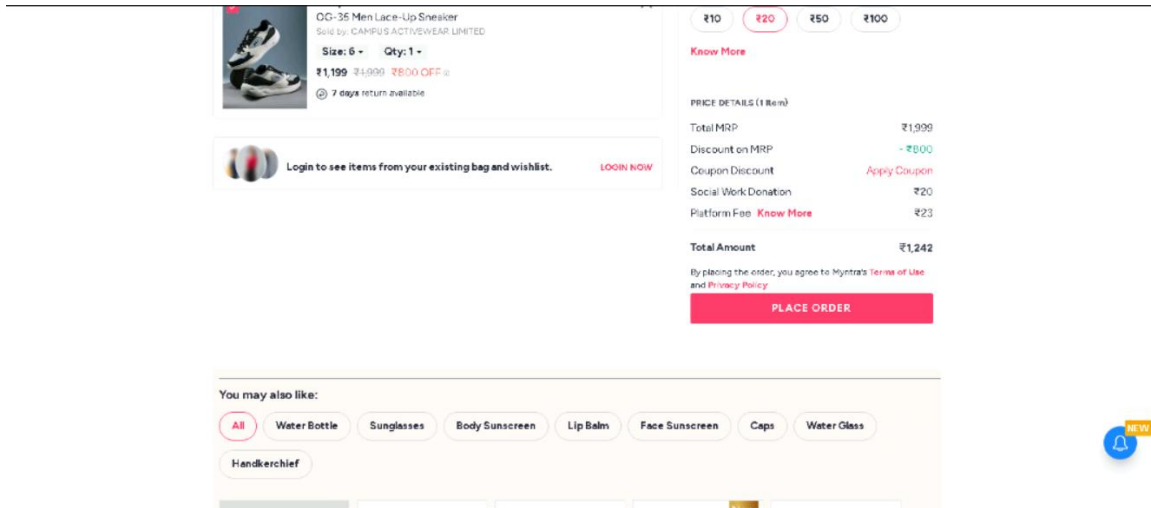
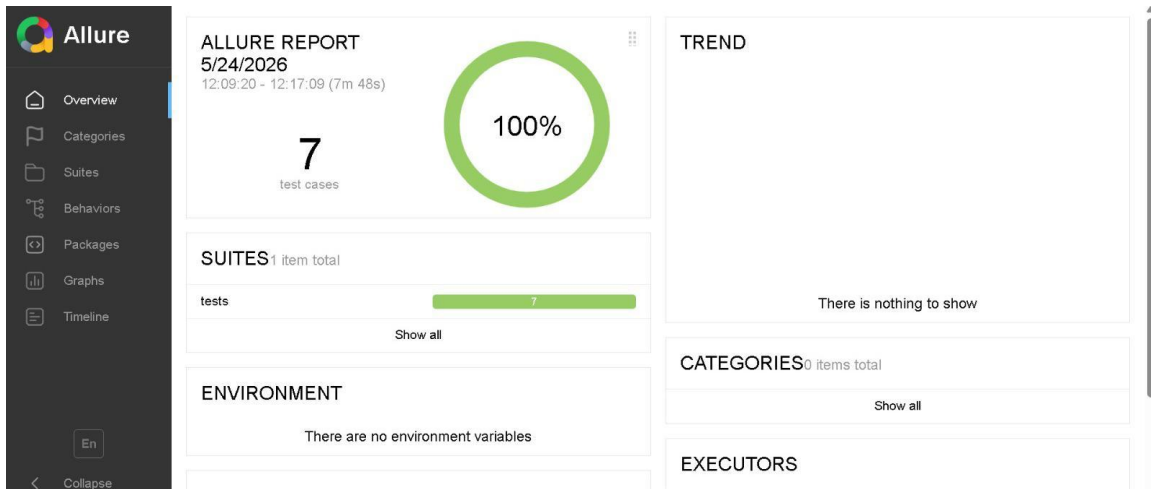
ALLURE REPORTS

- Integrated Allure Reports with Behave BDD Framework
- Generates professional execution reports
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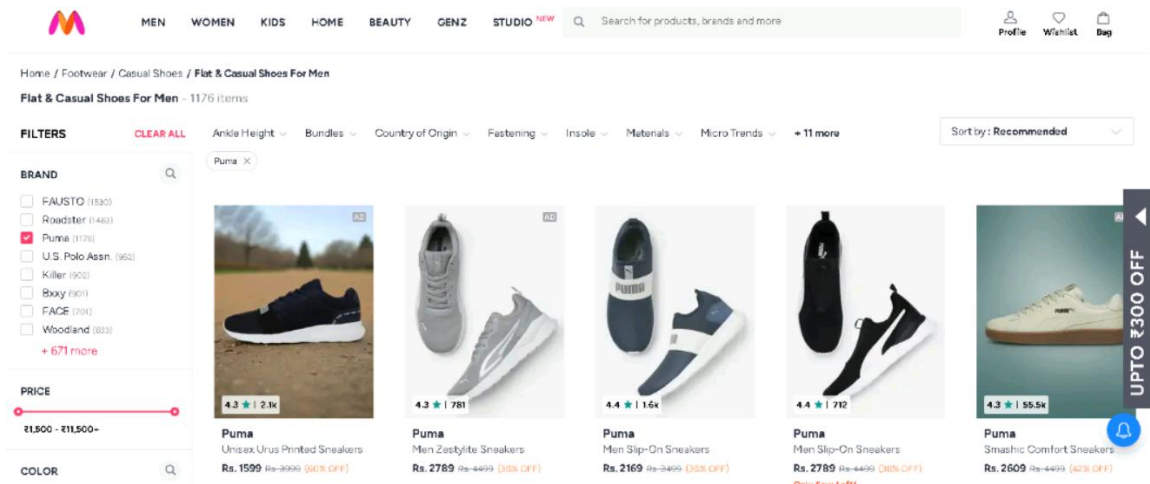
TEST OUTPUTS

The framework automatically captures screenshots during important execution stages including product selection page, cart page, wishlist popup, filter application page, and login verification page.

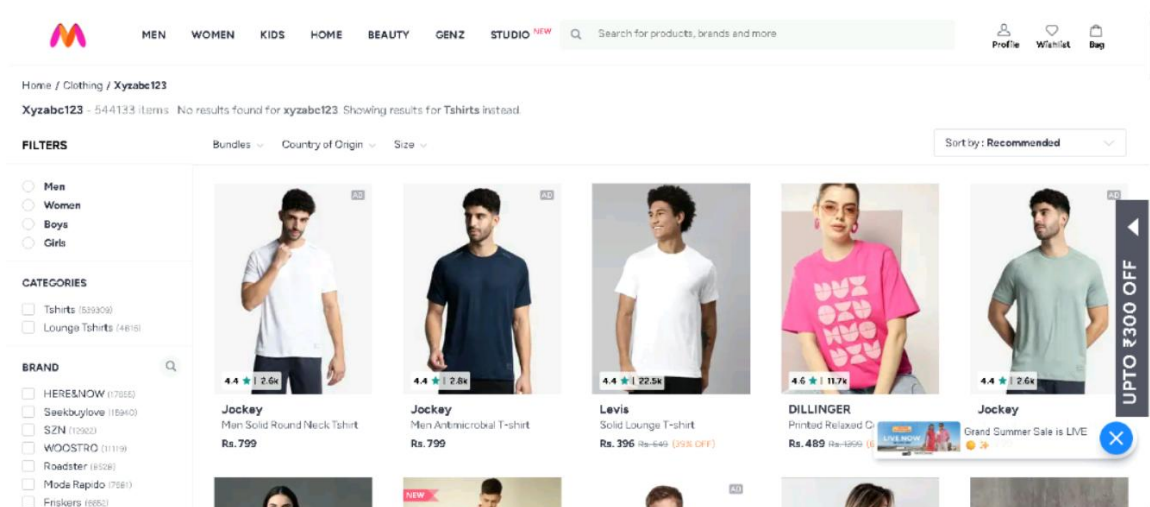
Detailed execution logs are generated using Python Logging framework for debugging and execution tracking.



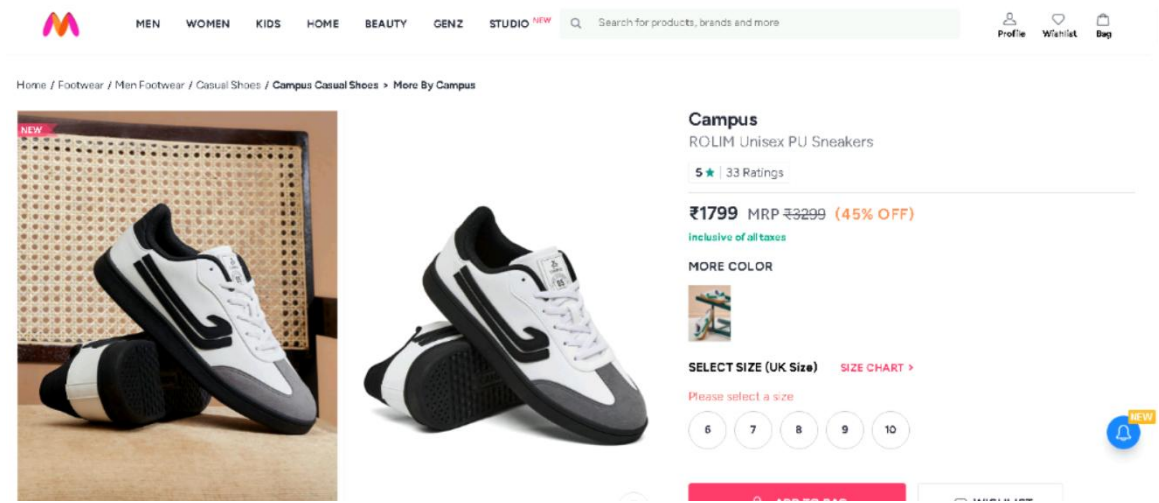
E2E Screenshot



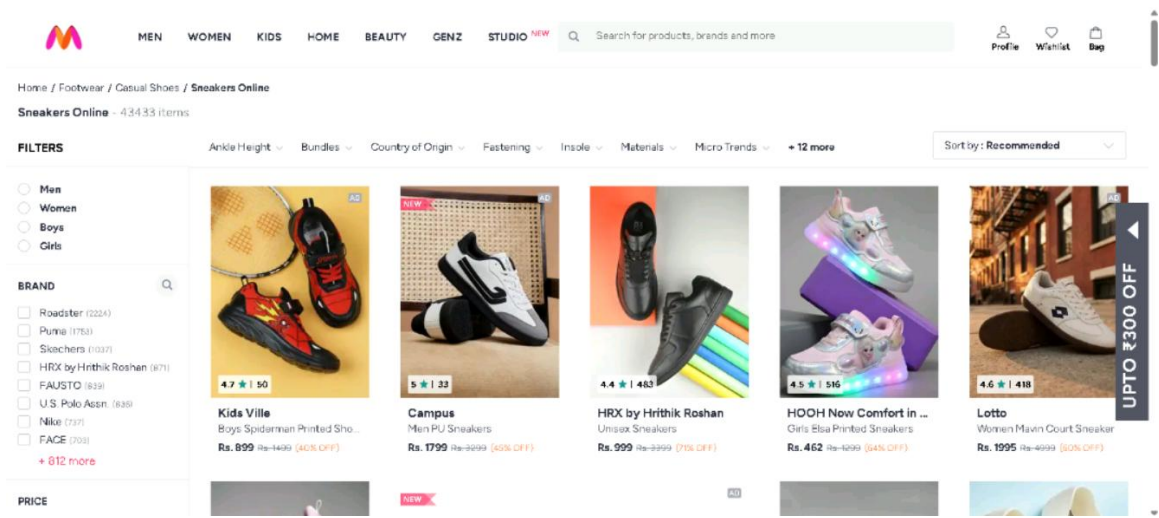
Filter Section



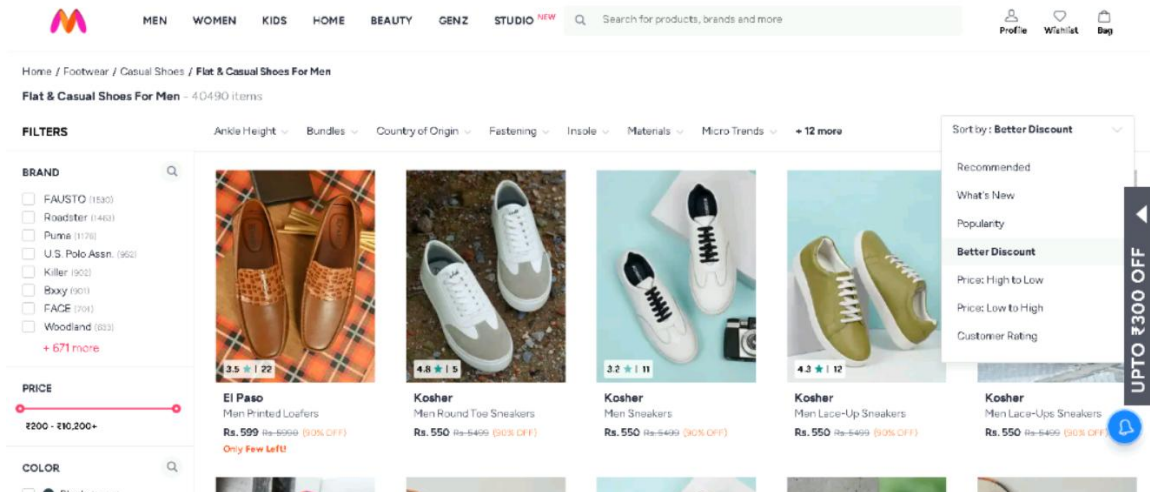
Invalid Search



No Size Selected



Search



Sort

TEST REPORT

- Total Scenarios Executed: 7
- Scenarios Passed: 7
- Scenarios Failed: 0
- Automation Accuracy: 100%

CONCLUSION

The Behave BDD automation framework successfully validates the Myntra MEN module using End-to-End, Positive, and Negative testing approaches.

The framework is designed using:

- Selenium WebDriver
- Behave BDD Framework
- Gherkin Syntax

- Page Object Model (POM)
- Allure Reporting
- Logging Framework

The automation framework improves:

- Readability
- Collaboration
- Maintainability
- Scalability
- Reporting Quality
- Automation Reliability

The framework successfully validates critical business workflows including MEN navigation, product search, filter validation, sorting validation, wishlist functionality, donation workflow, and place order verification.

All scenarios executed successfully with professional reporting, screenshot capture, and detailed execution logs.

Overall, the project demonstrates a scalable and professional BDD automation framework for real-world e-commerce application testing